

**SUPPLEMENTARY MATERIAL FOR:
SURVEY-WEIGHTED ORDINAL CAUSAL DISCOVERY: APPLICATION TO
VETERAN SOCIAL NEEDS AND HEALTH INDICATORS**

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This supplement provides: §1 the `swa-oBn` algorithm in pseudocode; §2 full proofs of all theoretical results stated in the main text; §3 details of the polychoric clustering procedure; §4 simulation Study 1 supplementary metrics (edge precision and correct orientation rate); §5 simulation Study 3 supplementary tables (paired SHD gaps across L -regimes and sample sizes, weighted and unweighted arms); §6 simulation Study 4 (survey-weighted PC, FCI, and LiNGAM); §7 proportional odds diagnostics; §8 survey weight distribution details; §9 the composite node table linking the analytic nodes to their constituent SHEP items; and §10 the complete 10×10 bootstrap edge probability matrix from the application. References to assumptions, propositions, and theorems refer to the main text unless otherwise noted.

1. The `swa-oBn` algorithm. The `swa-oBn` procedure is parameterized by the scoring choice `score_type` $\in \{\text{SINGLE-BIC}, \text{SPLIT-BIC}\}$ defined in Section 4 of the main text. Algorithm 1 formalizes the move-evaluation function used inside the greedy search; Algorithm 2 formalizes the full procedure.

Algorithm 1 SCORE_OBN: score-difference function for a candidate move

Require: Current DAG $\mathcal{G}$; candidate move $m \in \{\text{add}, \text{delete}, \text{reverse}\}$; data $(\mathbf{X}, \mathbf{w}, \mathbf{Z})$; effective sample size n_{eff} ; scoring choice `score_type`

Ensure: Score difference Δ for applying m to $\mathcal{G}$

- 1: Let $\mathcal{G}'$ denote the DAG obtained by applying m to $\mathcal{G}$
 - 2: **if** `score_type` = SPLIT-BIC **and** m = reverse **then**
 - 3: $\Delta \leftarrow \frac{n_{\text{eff}}}{n} \left[\tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w | \mathcal{G}') - \tilde{\ell}(\hat{\theta}_{\mathcal{G}}^w | \mathcal{G}) \right]$ {Rescaled unpenalized pseudo-log-likelihood (cf. Section 4.2 of main text)}
 - 4: **else**
 - 5: $\Delta \leftarrow -\frac{1}{2} \left[\widehat{\text{BIC}}(\mathcal{G}') - \widehat{\text{BIC}}(\mathcal{G}) \right]$ {Pseudo-BIC; cf. Equation 6 of main text}
 - 6: **end if**
 - 7: **return** Δ
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2. Proofs of Theoretical Results. Throughout, we use the notation established in Section 5 of the main text: ξ denotes the superpopulation, U_N the finite population of size N , S_n the probability sample of size n , $w_i = 1/\pi_i$ the survey weights, and $P_{\xi}(\mathbf{X} | \mathcal{G}, \theta, \mathbf{Z})$ the ordinal BN model under DAG $\mathcal{G}$ with parameters θ and covariates $\mathbf{Z}$. We define:

- Census log-likelihood: $\ell_U(\theta | \mathcal{G}) = \sum_{i \in U_N} \log P(\mathbf{x}_i | \mathcal{G}, \theta, \mathbf{z}_i)$.

Algorithm 2 swa-oBn: Survey-weighted, covariate-adjusted ordinal causal discovery

Require: Ordinal data $\mathbf{X} \in \mathbb{R}^{n \times q}$; survey weights $\mathbf{w} = (w_1, \dots, w_n)$ (optional); covariates $\mathbf{Z} \in \mathbb{R}^{n \times r}$ (optional); blacklist $\mathcal{B}$, whitelist $\mathcal{W}$ (optional); number of restarts R ; number of bootstrap replicates B ; scoring choice $\text{score_type} \in \{\text{SINGLE-BIC}, \text{SPLIT-BIC}\}$

Ensure: Point-estimate DAG $\hat{\mathcal{G}}$; edge inclusion probabilities $\{\hat{p}_{jk}\}$

{Preprocessing}
 1: Normalize weights: $w_i \leftarrow w_i \cdot n / \sum_{i'} w_{i'}$
 2: Compute $n_{\text{eff}} \leftarrow (\sum_i w_i)^2 / \sum_i w_i^2$
{Point estimate via multi-start hill-climbing}
 3: **for** $s = 1, \dots, R$ **do**
 4: Initialize $\mathcal{G}^{(s)}$ (empty graph if $s = 1$; random sparse DAG otherwise)
 5: Enforce whitelist edges in $\mathcal{G}^{(s)}$
 6: **repeat**
 7: For each candidate move m consistent with $\mathcal{B}$, $\mathcal{W}$, and acyclicity, compute $\Delta_m \leftarrow \text{SCORE_OBN}(\mathcal{G}^{(s)}, m, \mathbf{X}, \mathbf{w}, \mathbf{Z}, n_{\text{eff}}, \text{score_type})$
 8: Apply the move with the largest $\Delta_m > 0$, if any exists
 9: **until** no improving move exists
 10: **end for**
 11: $\hat{\mathcal{G}} \leftarrow \arg \min_s \widehat{\text{BIC}}(\mathcal{G}^{(s)})$
{Bootstrap uncertainty quantification}
 12: **for** $b = 1, \dots, B$ **do**
 13: Draw indices $\mathcal{I}^{(b)}$ by sampling $\{1, \dots, n\}$ with replacement
 14: Construct $(\mathbf{X}^{(b)}, \mathbf{w}^{(b)}, \mathbf{Z}^{(b)})$ from $\mathcal{I}^{(b)}$; recompute $n_{\text{eff}}^{(b)}$
 15: Run lines 3–10 on $(\mathbf{X}^{(b)}, \mathbf{w}^{(b)}, \mathbf{Z}^{(b)})$ to obtain $\hat{\mathcal{G}}^{(b)}$
 16: **end for**
 17: $\hat{p}_{jk} \leftarrow B^{-1} \sum_{b=1}^B \mathbf{1}(j \rightarrow k \in \hat{\mathcal{G}}^{(b)})$ for all (j, k)

- Pseudo-log-likelihood: $\tilde{\ell}(\theta \mid \mathcal{G}) = \sum_{i \in S_n} w_i \log P(\mathbf{x}_i \mid \mathcal{G}, \theta, \mathbf{z}_i)$, with raw weights $w_i = 1/\pi_i$. The normalized-weight form used in Equation (6) of the main text—which applies the rescaling (n_{eff}/n) to the pseudo-log-likelihood—differs from $\tilde{\ell}$ by a graph-invariant factor and yields the same arg min; see the pseudo-BIC parameterization remark in main §5.2.
- Superpopulation expected log-likelihood: $Q_\xi(\theta \mid \mathcal{G}) = \mathbb{E}_\xi[\log P(\mathbf{X} \mid \mathcal{G}, \theta, \mathbf{Z})]$.
- Normalized versions: $\bar{\ell}_U(\theta \mid \mathcal{G}) = N^{-1} \ell_U(\theta \mid \mathcal{G})$, $\bar{\tilde{\ell}}(\theta \mid \mathcal{G}) = N^{-1} \tilde{\ell}(\theta \mid \mathcal{G})$.

We use $\theta_{\mathcal{G}}^*$ to denote the superpopulation maximizer of $Q_\xi(\cdot \mid \mathcal{G})$, i.e., $\theta_{\mathcal{G}}^* = \arg \max_\theta Q_\xi(\theta \mid \mathcal{G})$; when $\mathcal{G} = \mathcal{G}^*$, this coincides with the true data-generating parameter, which we denote simply by θ^* . For a misspecified DAG $\mathcal{G}' \neq \mathcal{G}^*$, we use $\theta_{\mathcal{G}'}^\dagger \equiv \theta_{\mathcal{G}'}^*$ to emphasize its role as the KL-projection rather than a truth-level parameter. The *conditional* (covariate-pointwise) KL-projection is denoted $\theta^\dagger(\mathbf{z}) = \arg \max_\theta \sum_{\mathbf{x}} P(\mathbf{x} \mid \mathcal{G}^*, \theta^*, \mathbf{z}) \log P(\mathbf{x} \mid \mathcal{G}', \theta, \mathbf{z})$, which in general differs from the unconditional $\theta_{\mathcal{G}'}^\dagger$.

2.1. Proof of Proposition 1 (Identifiability under Covariate Adjustment). Let $\mathcal{G}^* \neq \mathcal{G}'$ be Markov-equivalent DAGs. Write $\vartheta = (\theta, \delta) \in \Theta \times \mathbb{R}^{r+q}$ for the full parameter under the covariate-adjusted model, and define the pointwise pseudo-true parameter

$$\theta^\dagger(\mathbf{z}; \vartheta^*) = \arg \max_\theta \sum_{\mathbf{x}} P(\mathbf{x} \mid \mathcal{G}^*, \vartheta^*, \mathbf{z}) \log P(\mathbf{x} \mid \mathcal{G}', \theta, \mathbf{z}).$$

Define the exceptional set

$$\mathcal{E} = \left\{ \vartheta^* : \mathbb{E}_{\mathbf{Z}} \left[\text{KL} \left(P_\xi(\mathbf{X} \mid \mathcal{G}^*, \vartheta^*, \mathbf{Z}) \parallel P(\mathbf{X} \mid \mathcal{G}', \theta^\dagger(\mathbf{Z}; \vartheta^*), \mathbf{Z}) \right) \right] = 0 \right\}.$$

We show that $\mathcal{E}$ has Lebesgue measure zero in $\Theta \times \mathbb{R}^{r+q}$.

2.1.1. *Step 1: Reduction to the no-covariate case.* For each fixed $\mathbf{Z} = \mathbf{z}$, the covariate-adjusted cumulative link model becomes

$$P_\xi(X_j \leq \ell \mid \mathbf{X}_{\text{pa}(j)}, \mathbf{Z} = \mathbf{z}) = F\left(\gamma_{j\ell} - \sum_{k \in \text{pa}(j)} \beta_{jk} X_k - \tilde{\alpha}_j(\mathbf{z})\right),$$

with $\tilde{\alpha}_j(\mathbf{z}) = \alpha_j + \boldsymbol{\delta}'_j \mathbf{z}$. This is a no-covariate oBN with thresholds $\{\gamma_{j\ell}\}$, parent-effect parameters $\{\beta_{jk}\}$, and shifted intercepts $\{\tilde{\alpha}_j(\mathbf{z})\}$.

2.1.2. *Step 2: The identifiability variety is intercept-invariant.* The identifiability proof of Ni and Mallick (2022) establishes that the ordinal parameterizations $P(X_j \mid X_k)$ and $P(X_k \mid X_j)$ are generically non-equivalent by analyzing a polynomial system in the threshold and parent-effect parameters. The intercept α_j determines the marginal location of the conditional distribution but does not enter the polynomial relations that govern distributional equivalence. Consequently, the exceptional set

$$\mathcal{V} = \left\{(\gamma, \beta, \alpha) : P(\mathbf{X} \mid \mathcal{G}^*, \gamma, \beta, \alpha) \equiv P(\mathbf{X} \mid \mathcal{G}', \theta_{\text{NC}}^\dagger(\gamma, \beta, \alpha))\right\}$$

in the no-covariate parameter space is a proper algebraic subvariety of Lebesgue measure zero, invariant under intercept translation. Here $\theta_{\text{NC}}^\dagger$ denotes the no-covariate pseudo-true parameter. By Assumption 5, $\mathcal{V}$ has Lebesgue measure zero.

2.1.3. *Step 3: The exceptional set has measure zero in $(\theta^*, \boldsymbol{\delta}^*)$ -space.* *Case (a): $\mathbf{Z}$ is continuous with a density on $\mathbb{R}^r$.* Fix $(\theta^*, \boldsymbol{\delta}^*)$ and consider the map $\mathbf{z} \mapsto (\gamma^*, \beta^*, \tilde{\alpha}^*(\mathbf{z}))$. This is an affine map. Since $\mathcal{V}$ has Lebesgue measure zero and the affine map has nonzero Jacobian for $\boldsymbol{\delta}^*$ outside a measure-zero degenerate set, the pre-image $\{\mathbf{z} : (\gamma^*, \beta^*, \tilde{\alpha}^*(\mathbf{z})) \in \mathcal{V}\}$ has Lebesgue measure zero in $\mathbf{z}$ -space. Therefore the pointwise KL is strictly positive for $P_{\mathbf{Z}}$ -almost every $\mathbf{z}$, and

$$\mathbb{E}_{\mathbf{Z}} \left[\text{KL} \left(P_\xi(\cdot \mid \mathcal{G}^*, \vartheta^*, \mathbf{Z}) \parallel P(\cdot \mid \mathcal{G}', \theta^\dagger(\mathbf{Z}; \vartheta^*), \mathbf{Z}) \right) \right] > 0$$

whenever $\boldsymbol{\delta}^*$ avoids this degenerate set. *Case (b): $\mathbf{Z}$ is discrete with finite support $\{\mathbf{z}_1, \dots, \mathbf{z}_K\}$.* For each k , define

$$\mathcal{E}_k = \left\{(\theta^*, \boldsymbol{\delta}^*) : (\gamma^*, \beta^*, \alpha_j^* + (\boldsymbol{\delta}_j^*)^\top \mathbf{z}_k) \in \mathcal{V}\right\}.$$

For each fixed $\mathbf{z}_k$, the map $(\theta^*, \boldsymbol{\delta}^*) \mapsto (\gamma^*, \beta^*, \alpha_j^* + (\boldsymbol{\delta}_j^*)^\top \mathbf{z}_k)$ is affine with nonzero Jacobian (provided $\mathbf{z}_k \neq \mathbf{0}$). By Fubini's theorem and the measure-zero property of $\mathcal{V}$, $\mathcal{E}_k$ has Lebesgue measure zero in $(\theta^*, \boldsymbol{\delta}^*)$ -space. The overall exceptional set satisfies $\mathcal{E} \subseteq \bigcup_{k=1}^K \mathcal{E}_k$, which, as a finite union of measure-zero sets, has Lebesgue measure zero. (When $\mathbf{z}_k = \mathbf{0}$ for some k , we redefine the origin so that all $\mathbf{z}_k$ are nonzero; this does not change the model.) Therefore $\mathcal{E}$ has Lebesgue measure zero, as claimed. When the true parameter $(\theta^*, \boldsymbol{\delta}^*) \notin \mathcal{E}$, the expected KL divergence is strictly positive. $\square$

2.2. *Proof of Proposition 2 (Pseudo-MLE Consistency).* We prove that $\hat{\theta}_{\mathcal{G}}^w \xrightarrow{p} \theta_{\mathcal{G}}^*$ under the joint $(\xi \times p_N)$ -probability. The argument proceeds in three steps.

2.2.1. *Step 1: Design consistency of the normalized pseudo-log-likelihood.* Fix $\theta \in \Theta_{\mathcal{G}}$ and define $g_{\theta}(\mathbf{x}, \mathbf{z}) = \log P(\mathbf{x} \mid \mathcal{G}, \theta, \mathbf{z})$. Since $\mathbf{X}$ takes values in the finite set $\prod_{j=1}^q \{1, \dots, L_j\}$, Assumption 6 (Non-Degeneracy) gives $P(\mathbf{x} \mid \mathcal{G}, \theta, \mathbf{z}) > 0$ for all $(\mathbf{x}, \mathbf{z})$, and compactness of $\Theta_{\mathcal{G}}$ (Assumption 10(i)) combined with continuity of F yields a uniform lower bound $P(\mathbf{x} \mid \mathcal{G}, \theta, \mathbf{z}) \geq p_{\min} > 0$ over all $(\mathbf{x}, \mathbf{z}, \theta)$. Therefore g_{θ} is bounded uniformly in $\theta \in \Theta_{\mathcal{G}}$. By Assumption 9 (Design Consistency), applied to the function g_{θ} of $(\mathbf{x}, \mathbf{z})$,

$$(1) \quad \bar{\ell}(\theta \mid \mathcal{G}) - \bar{\ell}_U(\theta \mid \mathcal{G}) = \frac{1}{N} \sum_{i \in S_n} w_i g_{\theta}(\mathbf{x}_i, \mathbf{z}_i) - \frac{1}{N} \sum_{i \in U_N} g_{\theta}(\mathbf{x}_i, \mathbf{z}_i) \xrightarrow{p_N} 0.$$

Design Consistency in turn follows from Horvitz–Thompson consistency under Assumptions 7–8 (Positivity, Bounded Weights); see [Fuller \(2009\)](#), Ch. 1.

2.2.2. *Step 2: Superpopulation law of large numbers for the census log-likelihood.* By the i.i.d. superpopulation structure, $\{(\mathbf{x}_i, \mathbf{z}_i)\}_{i \in U_N}$ are i.i.d. draws from ξ . Since g_{θ} is bounded, the strong law of large numbers gives

$$(2) \quad \bar{\ell}_U(\theta \mid \mathcal{G}) = \frac{1}{N} \sum_{i=1}^N g_{\theta}(\mathbf{x}_i, \mathbf{z}_i) \xrightarrow{\text{a.s.}} \mathbb{E}_{\xi}[g_{\theta}(\mathbf{X}, \mathbf{Z})] = Q_{\xi}(\theta \mid \mathcal{G}).$$

2.2.3. *Step 3: Uniform convergence and argmax convergence.* Combining (1) and (2), for each fixed θ : $\bar{\ell}(\theta \mid \mathcal{G}) \xrightarrow{p} Q_{\xi}(\theta \mid \mathcal{G})$. We upgrade this to uniform convergence over $\Theta_{\mathcal{G}}$. By Assumption 10(iv), for any $\theta, \theta' \in \Theta_{\mathcal{G}}$,

$$|g_{\theta}(\mathbf{x}, \mathbf{z}) - g_{\theta'}(\mathbf{x}, \mathbf{z})| \leq M \|\theta - \theta'\|$$

for some constant $M > 0$. This Lipschitz condition, combined with the compactness of $\Theta_{\mathcal{G}}$ (Assumption 10(i)), gives equicontinuity. Pointwise convergence plus equicontinuity on a compact set gives uniform convergence ([van der Vaart, 1998](#), Theorem 2.4):

$$(3) \quad \sup_{\theta \in \Theta_{\mathcal{G}}} \left| \bar{\ell}(\theta \mid \mathcal{G}) - Q_{\xi}(\theta \mid \mathcal{G}) \right| \xrightarrow{p} 0.$$

By Assumption 10(iii), $Q_{\xi}(\theta \mid \mathcal{G})$ has a unique maximizer $\theta_{\mathcal{G}}^*$ with nonsingular Hessian. By (3) and the argmax continuous mapping theorem ([van der Vaart, 1998](#), Theorem 5.7),

$$(4) \quad \hat{\theta}_{\mathcal{G}}^w \xrightarrow{p} \theta_{\mathcal{G}}^*.$$

As a corollary:

$$(5) \quad \bar{\ell}(\hat{\theta}_{\mathcal{G}}^w \mid \mathcal{G}) \xrightarrow{p} Q_{\xi}(\theta_{\mathcal{G}}^* \mid \mathcal{G}).$$

□

2.3. *Proof of Theorem 5.1 (Skeleton Consistency).* Let $\mathcal{G}^*$ be the true DAG and $\mathcal{G}'$ a DAG not Markov-equivalent to $\mathcal{G}^*$. We show $\Pr(\widehat{\text{BIC}}(\mathcal{G}^*) < \widehat{\text{BIC}}(\mathcal{G}')) \rightarrow 1$. We write

$$(6) \quad \widehat{\text{BIC}}(\mathcal{G}^*) - \widehat{\text{BIC}}(\mathcal{G}') = -2 \left[\bar{\ell}(\hat{\theta}_{\mathcal{G}^*}^w \mid \mathcal{G}^*) - \bar{\ell}(\hat{\theta}_{\mathcal{G}'}^w \mid \mathcal{G}') \right] + (K_{\mathcal{G}^*} - K_{\mathcal{G}'})\rho_n.$$

2.3.1. *Case 1A: $\mathcal{G}'$ is an overfit model ($\mathcal{G}^* \subset \mathcal{G}'$ as directed graphs, so $K_{\mathcal{G}'} > K_{\mathcal{G}^*}$).* Since $\mathcal{G}^*$ is a subgraph of $\mathcal{G}'$, the oBN model under $\mathcal{G}^*$ is nested within the model under $\mathcal{G}'$, and both achieve the same superpopulation optimum: $Q_\xi(\theta_{\mathcal{G}^*}^* | \mathcal{G}^*) = Q_\xi(\theta_{\mathcal{G}'}^\dagger | \mathcal{G}')$. By the nested-model likelihood-ratio theory for pseudo-maximum likelihood (Rao and Scott, 1981; Lumley and Scott, 2014), the survey-weighted pseudo-log-likelihood ratio is

$$2 \left[\tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w | \mathcal{G}') - \tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w | \mathcal{G}^*) \right] = O_p(1),$$

asymptotically distributed as a weighted mixture of chi-squared random variables with weights determined by the design effects of the added parameters. Meanwhile, $(K_{\mathcal{G}^*} - K_{\mathcal{G}'})\rho_n < 0$ with $|K_{\mathcal{G}^*} - K_{\mathcal{G}'}|\rho_n \rightarrow \infty$ since $\rho_n \rightarrow \infty$. Therefore

$$\widetilde{\text{BIC}}(\mathcal{G}^*) - \widetilde{\text{BIC}}(\mathcal{G}') = O_p(1) + (K_{\mathcal{G}^*} - K_{\mathcal{G}'})\rho_n \rightarrow -\infty$$

in probability.

2.3.2. *Case 1B: $\mathcal{G}'$ has more edges than $\mathcal{G}^*$ but is not a supergraph.* In this case, $\mathcal{G}'$ is misspecified: it both adds spurious edges and omits at least one true edge. The omission places $\mathcal{G}'$ in the regime of Case 2 below, where the likelihood gap dominates the penalty difference regardless of sign.

2.3.3. *Case 2: $\mathcal{G}'$ is an underfit model (missing edges from $\mathcal{G}^*$).* By Assumption 3 (Faithfulness), every edge in $\mathcal{G}^*$ corresponds to a genuine conditional dependence: $Q_\xi(\theta^* | \mathcal{G}^*) > Q_\xi(\theta^\dagger | \mathcal{G}')$. Let $\Delta = Q_\xi(\theta^* | \mathcal{G}^*) - Q_\xi(\theta^\dagger | \mathcal{G}') > 0$. By (5), the likelihood term is $N(\Delta + o_p(1))$, dominating the $O(\rho_n)$ penalty since $\rho_n/N \rightarrow 0$:

$$\widetilde{\text{BIC}}(\mathcal{G}^*) - \widetilde{\text{BIC}}(\mathcal{G}') = -2N(\Delta + o_p(1)) + O(\rho_n) \rightarrow -\infty.$$

2.3.4. *Case 3: Same skeleton, not Markov-equivalent.* This can only occur if $\mathcal{G}'$ and $\mathcal{G}^*$ differ in v-structures. By Assumption 3, $\mathcal{G}'$ implies at least one conditional independence not present in P_ξ , making it effectively misspecified (as in Case 2). $\square$

2.4. *Proof of Theorem 5.2 (Orientation Consistency).* Let $\mathcal{G}^*$ be the true DAG and $\mathcal{G}'$ a Markov-equivalent DAG with $\mathcal{G}' \neq \mathcal{G}^*$. We show $\Pr(\tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w | \mathcal{G}^*) > \tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w | \mathcal{G}')) \rightarrow 1$.

2.4.1. *Step 1: Convergence of the pseudo-log-likelihood ratio.* Applying (5) to both $\mathcal{G}^*$ and $\mathcal{G}'$:

$$(7) \quad \frac{1}{N} \left[\tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w | \mathcal{G}^*) - \tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w | \mathcal{G}') \right] \xrightarrow{p} Q_\xi(\theta_{\mathcal{G}^*}^* | \mathcal{G}^*) - Q_\xi(\theta_{\mathcal{G}'}^\dagger | \mathcal{G}'),$$

where $\theta_{\mathcal{G}^*}^* = \theta^*$ (since $\mathcal{G}^*$ is correctly specified) and $\theta_{\mathcal{G}'}^\dagger = \arg \max_\theta Q_\xi(\theta | \mathcal{G}')$ is the superpopulation KL-projection onto the model under $\mathcal{G}'$.

2.4.2. *Step 2: Identification of the limit as an expected KL divergence.* Expanding the definition of Q_ξ :

$$\begin{aligned} Q_\xi(\theta^* | \mathcal{G}^*) - Q_\xi(\theta_{\mathcal{G}'}^\dagger | \mathcal{G}') &= \mathbb{E}_\xi \left[\log P(\mathbf{X} | \mathcal{G}^*, \theta^*, \mathbf{Z}) - \log P(\mathbf{X} | \mathcal{G}', \theta_{\mathcal{G}'}^\dagger, \mathbf{Z}) \right] \\ &= \mathbb{E}_\mathbf{Z} \left\{ \mathbb{E}_\xi \left[\log \frac{P(\mathbf{X} | \mathcal{G}^*, \theta^*, \mathbf{Z})}{P(\mathbf{X} | \mathcal{G}', \theta_{\mathcal{G}'}^\dagger, \mathbf{Z})} \middle| \mathbf{Z} \right] \right\} \\ &\geq \mathbb{E}_\mathbf{Z} \left\{ \mathbb{E}_\xi \left[\log \frac{P(\mathbf{X} | \mathcal{G}^*, \theta^*, \mathbf{Z})}{P(\mathbf{X} | \mathcal{G}', \theta^\dagger(\mathbf{Z}), \mathbf{Z})} \middle| \mathbf{Z} \right] \right\} \\ &= \mathbb{E}_\mathbf{Z} \left[\text{KL} \left(P_\xi(\mathbf{X} | \mathcal{G}^*, \theta^*, \mathbf{Z}) \parallel P(\mathbf{X} | \mathcal{G}', \theta^\dagger(\mathbf{Z}), \mathbf{Z}) \right) \right], \end{aligned} \quad (8)$$

where the inequality in (8) holds because $\theta^\dagger(z)$ is the pointwise KL-minimizer given $\mathbf{Z} = z$, so each inner expectation is at least as large as when evaluated at any other parameter. By Proposition 1, the final expected KL is strictly positive under the generic-parameter assumption on (θ^*, δ^*) : let $\Delta > 0$ denote its value.

2.4.3. *Step 3: Conclusion.* The pseudo-log-likelihood difference is $N[\Delta + o_p(1)] \rightarrow +\infty$ in probability, establishing the result. $\square$

REMARK 1. *No penalty term appears in this proof. The orientation comparison is a pseudo-likelihood ratio, and its consistency follows from generic ordinal identifiability (Proposition 1) and pseudo-MLE consistency (Proposition 2). Both scoring variants of Section 4 inherit this consistency: split-BIC compares reversals on the pseudo-likelihood directly, while single-BIC adds a penalty differential of order $O(\rho_n)$ that is asymptotically dominated by the $O(N)$ likelihood gap. The variants differ only in finite-sample behavior under heterogeneous $\mathbf{L}$; see Section 7.*

3. Polychoric Clustering Details. Figure 1 presents the weighted polychoric correlation matrix and hierarchical clustering dendrogram for the 13 SHEP social need items. Items within each highlighted cluster have pairwise polychoric correlations $\rho \geq 0.75$.

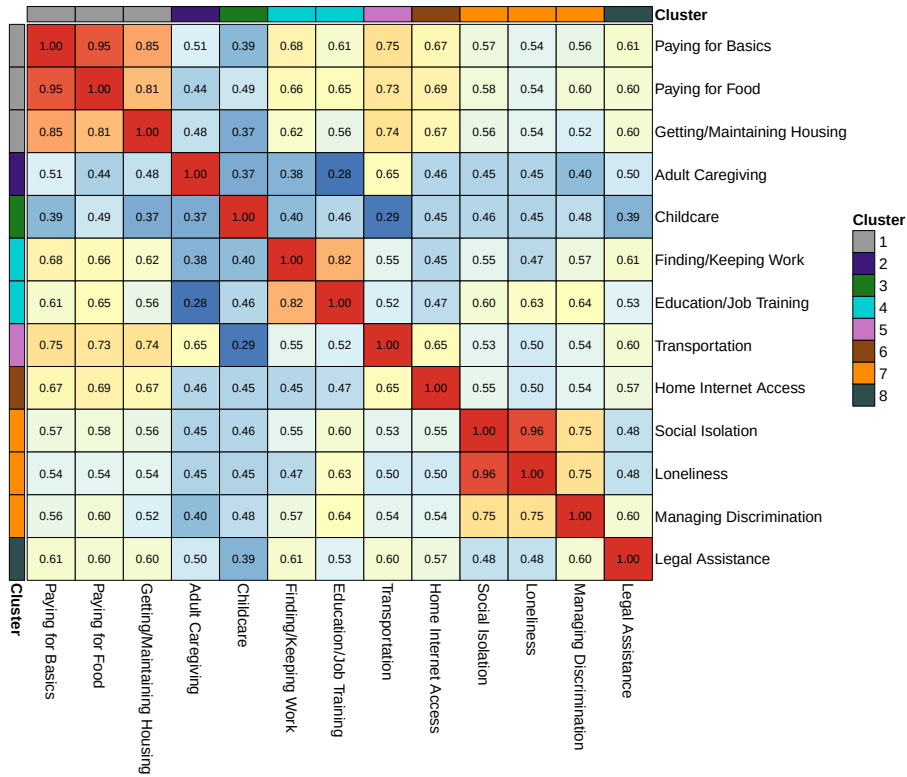


FIG 1. Polychoric correlation matrix and hierarchical clustering of 13 SHEP social need items. Shaded blocks indicate clusters with $\rho \geq 0.75$: basic needs (food, housing, paying for basics), work/education, and isolation/loneliness/discrimination.

4. Simulation Study 1: Edge Precision and Orientation Accuracy. Figure 2 presents edge precision (True Positive/(True Positive + False Positive)) for the swa-oBN versus naive oBN across design strength and signal strength conditions from Study 1 (Section 7.2 of the main text). Under mild and strong designs, the swa-oBN achieves substantially higher precision than the naive oBN (approximately 0.90–0.95 versus 0.70–0.75), indicating that the primary benefit of survey weighting is the elimination of spurious edges. Under no design, precision is identical for both methods.

Figure 3 presents the correct orientation rate (fraction of true edges found and correctly oriented). The weighted estimator records modestly lower orientation rates than the naive estimator under informative sampling, reflecting a precision-recall tradeoff: by suppressing spurious edges (Figure 2), swa-oBN also yields fewer spuriously oriented edges. The combined effect on SHD is favorable, as documented in Section 7.2 of the main text.

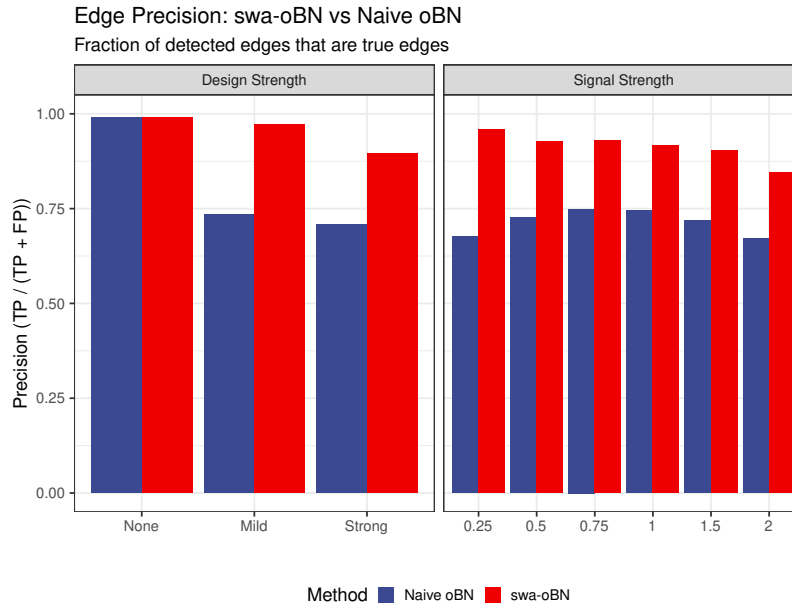


FIG 2. Edge precision ($TP/(TP+FP)$) of swa-oBN versus naive oBN across design strength (left) and signal strength (right) conditions. $q = 8$ nodes, 100 replications.

5. Simulation Study 3 Supplementary Tables. This appendix reports the full 5×4 grid of paired SHD gaps (split-BIC – single-BIC) underlying the headline result of Section 7.4. Tables 1 and 2 cover the weighted and unweighted arms respectively. Negative values indicate that split-BIC achieves lower SHD; significance is assessed at $|\text{gap}| > 1.96 \cdot \text{SE}$, with significant cells marked *.

Split-BIC dominates single-BIC in every regime at every sample size, with gap magnitudes growing in n , consistent with the per-edge bias of single-BIC’s penalty differential accumulating with effective sample size. The unweighted arm (Table 2) exhibits the same pattern with slightly larger gap magnitudes. The advantage of split-BIC over single-BIC scoring thus depends on the L -distribution alone, independent of survey weighting; the comparison of Section 7.4 applies to both swa-oBN and the unweighted oBN of Ni, Chen and Wang (2025).

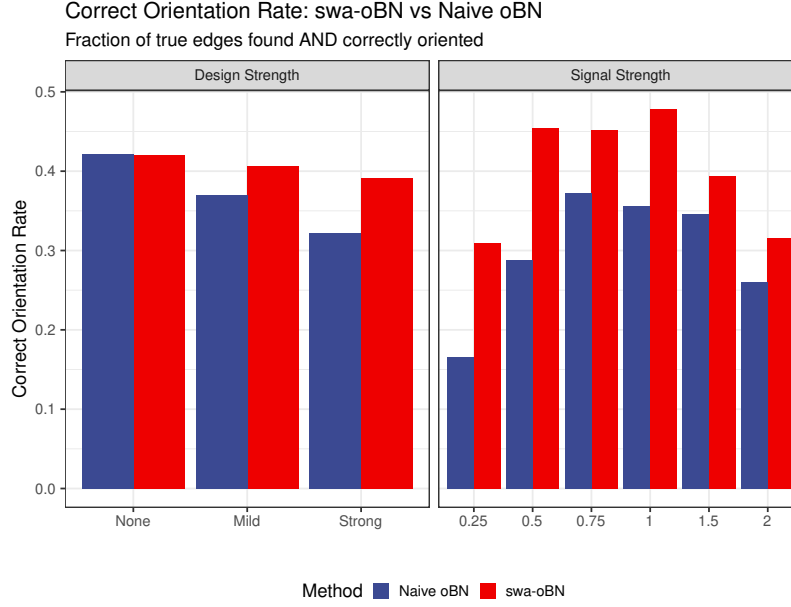


FIG 3. Correct orientation rate (fraction of true edges found and correctly oriented) across design strength (left) and signal strength (right) conditions. $q = 8$ nodes, 100 replications.

TABLE 1

Study 3, weighted arm: paired SHD gap (split-BIC – single-BIC) with standard errors across five $\mathbf{L}$ -regimes and four sample sizes. Negative values favor split-BIC; * denotes $|\text{gap}| > 1.96 \cdot \text{SE}$.

Regime	$\mathbf{L}$ -set	$n = 200$	$n = 400$	$n = 700$	$n = 1,000$
Uniform	(3, 3, 3, 3, 3, 3, 3, 3)	-0.07 (0.03)*	-0.05 (0.03)	-0.13 (0.06)*	-0.07 (0.02)*
Adjacent	(3, 3, 3, 3, 3, 4, 4, 4)	-0.23 (0.05)*	-0.23 (0.06)*	-0.28 (0.06)*	-0.46 (0.06)*
Polarized	(3, 3, 3, 3, 5, 5, 5, 5)	-0.21 (0.04)*	-0.24 (0.05)*	-0.36 (0.05)*	-0.61 (0.05)*
Wide	(3, 3, 3, 3, 3, 3, 3, 7)	-0.19 (0.04)*	-0.19 (0.04)*	-0.61 (0.05)*	-0.98 (0.07)*
Extreme	(3, 3, 3, 3, 3, 3, 3, 9)	-0.20 (0.04)*	-0.31 (0.05)*	-0.52 (0.05)*	-0.81 (0.06)*

TABLE 2

Study 3, unweighted arm: paired SHD gap (split-BIC – single-BIC) with standard errors across five $\mathbf{L}$ -regimes and four sample sizes. Negative values favor split-BIC; * denotes $|\text{gap}| > 1.96 \cdot \text{SE}$.

Regime	$\mathbf{L}$ -set	$n = 200$	$n = 400$	$n = 700$	$n = 1,000$
Uniform	(3, 3, 3, 3, 3, 3, 3, 3)	-0.08 (0.04)	-0.15 (0.04)*	-0.05 (0.04)	-0.16 (0.03)*
Adjacent	(3, 3, 3, 3, 3, 4, 4, 4)	-0.27 (0.05)*	-0.38 (0.06)*	-0.50 (0.06)*	-0.75 (0.05)*
Polarized	(3, 3, 3, 3, 5, 5, 5, 5)	-0.19 (0.06)*	-0.52 (0.05)*	-0.71 (0.05)*	-0.74 (0.05)*
Wide	(3, 3, 3, 3, 3, 3, 3, 7)	-0.31 (0.04)*	-0.49 (0.05)*	-1.03 (0.05)*	-1.06 (0.06)*
Extreme	(3, 3, 3, 3, 3, 3, 3, 9)	-0.19 (0.05)*	-0.55 (0.04)*	-0.65 (0.06)*	-1.03 (0.06)*

6. Simulation Study 4: Survey Weighting for Other Algorithm Families. To demonstrate that the distortion introduced by informative sampling is not specific to ordinal Bayesian networks, we evaluated survey-weighted and unweighted versions of three additional causal discovery algorithms: the PC algorithm (Spirtes, Glymour and Scheines, 2000), the FCI algorithm (Spirtes, Glymour and Scheines, 2000), and LiNGAM (Shimizu et al., 2006).

6.1. *Setup.* We generated a population of $N = 20,000$ units from a three-variable linear chain $X \rightarrow Y \rightarrow Z$ with continuous variables ($X, Z \sim N(0, 1)$, $Y = 0.8X + \varepsilon_Y$, $Z = 0.8Y + \varepsilon_Z$, $\varepsilon_Y, \varepsilon_Z \sim N(0, 1)$). From this population, Bernoulli samples were drawn under three design conditions (none, mild, strong), with oversampling based on the top quartile of the combined score of X and Z . Each scenario was replicated 100 times.

For the PC and FCI algorithms, survey weighting was implemented by replacing standard conditional independence tests with design-adjusted Wald tests using the `survey` package in R (Lumley, 2004). For LiNGAM, we implemented a weighted pseudo-population resampling procedure that restored i.i.d. structure by resampling from the observed data proportional to survey weights prior to applying the standard ICA-based algorithm.

6.2. *Results.* Figure 4 presents the correct skeleton recovery rate across the three design conditions for all three algorithm families. Under simple random sampling (design = none), all six estimators (three algorithms $\times$ two weighting conditions) recover the correct skeleton in nearly all replications. Under mild and strong designs, the unweighted versions show a marked decline in skeleton recovery, while the survey-weighted versions maintain substantially higher correct recovery rates.

Figure 5 presents the spurious edge rate (proportion of replications in which a direct $X-Z$ edge is inferred). Under no design, spurious edges are rare for all methods. Under mild and strong designs, the unweighted versions infer the spurious $X-Z$ edge at substantially elevated rates relative to their weighted counterparts. The effect is largest for FCI (95% spurious-edge rate unweighted vs. 4% weighted under strong design), substantial for the PC algorithm, and present but smaller for LiNGAM, whose ICA-based identification is less sensitive to design-induced marginal dependence.

These results confirm that the survey-weighting problem documented in Study 1 for the ordinal Bayesian network is a general phenomenon affecting all three major families of causal discovery algorithms.

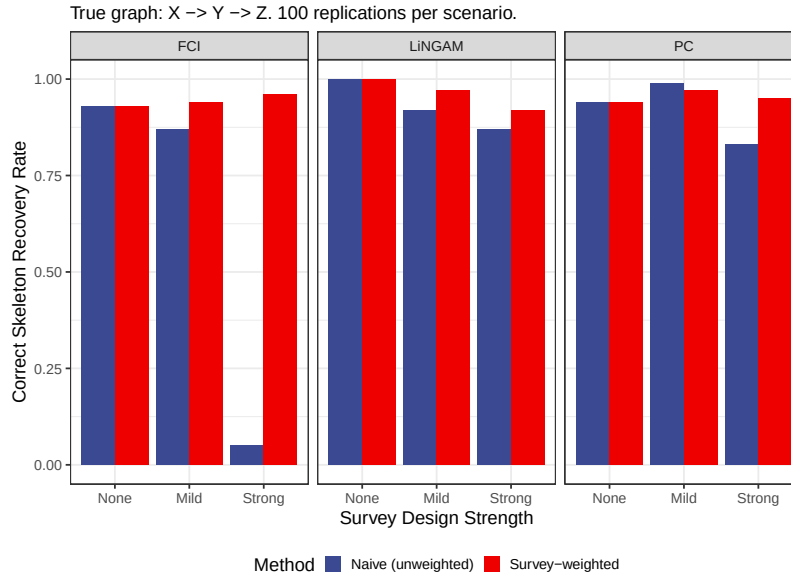


FIG 4. Study 4: Correct skeleton recovery rate for survey-weighted and unweighted versions of PC, FCI, and LiNGAM across three design conditions. True graph: $X \rightarrow Y \rightarrow Z$. 100 replications per scenario.

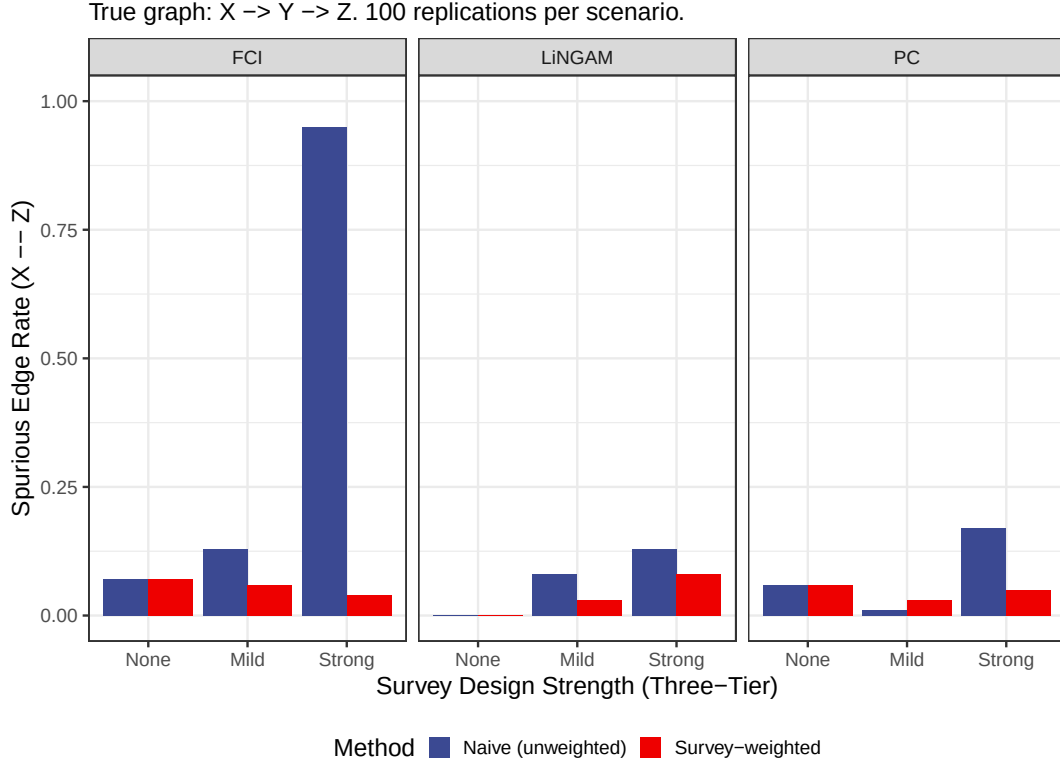


FIG 5. Study 4: Spurious X - Z edge rate for survey-weighted and unweighted versions of PC, FCI, and LiNGAM across three design conditions. True graph: $X \rightarrow Y \rightarrow Z$. 100 replications per scenario.

7. Proportional Odds Diagnostics. The `swa-obn` framework rests on the cumulative link specification of Assumption 4 of the main text, of which the proportional odds (PO) constraint is the most consequential working restriction: a single set of slope parameters is shared across the $L_j - 1$ cumulative thresholds for node j . This appendix reports diagnostic checks of PO for the application of Section 6 of the main text, documents the limitations of off-the-shelf diagnostics under a complex survey design, and presents a design-based sensitivity check for the regression whose orientation most directly drives our substantive conclusions.

7.1. Limitations of off-the-shelf diagnostics under a complex survey. The standard PO diagnostic is the [Brant \(1990\)](#) score test, implemented in R via `brant : : brant`. The test fits the cumulative link model via maximum likelihood and rejects PO if the between-threshold slope differences are jointly too large relative to the estimated covariance. In survey settings this procedure has two well-known failings. First, the estimator is a pseudo-MLE that treats observations as independently and identically distributed; the Hessian used to construct the score-test variance therefore omits the between-stratum and unequal-probability components of design variance and is anticonservative. Under design effect $deff$, the nominal test statistic is inflated by a factor of approximately $deff$, and at our sample size ($n_{eff} = 1,387$, $deff = 3.89$) this inflation can transform a modest PO departure into an apparently decisive rejection. Second, all B bootstrap replicates of the `swa-obn` refit cumulative link regressions at parent sets that are themselves estimated; the nominal Brant distribution does not account for this parent-set stochasticity. We therefore present nominal Brant results as indicative rather than confirmatory, and follow them with a design-based analog for the regression in the discovered DAG whose orientation drives our principal finding.

7.2. Nominal Brant results. We report Brant tests under two parent specifications. *Pre-analysis* regressed each of the $q = 10$ ordinal nodes on the demographic covariates $\mathbf{Z}$ alone (age band and race-ethnicity-sex), a specification that makes no use of the discovered DAG. *Post-analysis* regressed each node on its estimated parent set from the consensus DAG (Figure 1 of the main text) together with $\mathbf{Z}$. At nominal $\alpha = 0.05$, the pre-analysis Brant test rejected PO in 1 of the 10 nodes; the post-analysis test rejected PO in 2 of 10. The increase in rejections from pre- to post-analysis is consistent with the Brant statistic inflating mechanically as the parent set grows (each added parent contributes $L_k - 1$ additional slopes to the joint test) and does not, on its own, indicate that the PO structure deteriorates conditional on the discovered parents. Given the anticonservatism documented in Section 7.1, we do not interpret these counts as evidence against the working model.

7.3. Working-model interpretation. Following Lumley and Scott (2015), we treated the PO cumulative link as an explicitly designated working model under complex-survey pseudolikelihood. The generic identifiability result invoked throughout this paper (Proposition 1 of the main text) is a statement about the ordinal parameterization itself, not about whether the data-generating mechanism exactly satisfies PO. Under the working-model view, the pseudo-MLE $\hat{\theta}^w$ consistently estimates the KL projection of the population distribution onto the PO model under the sampling design; the bootstrap inclusion probabilities $\hat{p}_{jk}$ describe the dependency structure of this projection, not of the unrestricted population. The substantive consequence is that the consensus DAG should be read as summarizing the dominant directed dependencies in the SHEP data under the ordinal working model, which is the claim that the theory of Section 5 of the main text supports.

8. Survey Weight Distribution. Table 3 summarizes the distribution of survey weights in the SHEP analytic sample. The design effect ($\text{DEFF} = n/n_{\text{eff}}$) quantifies the loss of effective sample size due to unequal weighting.

TABLE 3
Survey weight distribution and design effect.

	n	n_{eff}	DEFF	Min	Median	Max	CV
Overall	5,396	1,387	3.89	3.4	58.4	3,330.1	1.700

9. Composite Node Construction. Table 4 maps the ten analytic nodes used in Section 6 of the main text to their constituent SHEP items, following the polychoric clustering of Appendix 3. Composites were constructed row-wise as the maximum reported severity across constituent items, preserving the three-level ordinal scale (no support needed; needed support and got it; needed support but did not get it). The resulting level vector $\mathbf{L} = (3, 3, 3, 3, 3, 3, 3, 3, 5, 4)$ is heterogeneous; per Section 7.4, the application used split-BIC scoring.

TABLE 4
Mapping of analytic nodes to constituent SHEP items. The first eight rows are need nodes; the last two are health outcomes.

Analytic node	Constituent SHEP items (Q43 subitem)	L_j	Construction
Caregiving	Obtaining adult caregiving for yourself or others	3	single
Childcare	Obtaining childcare	3	single
Transportation	Getting transportation for basic needs	3	single
Internet	Accessing the internet at home	3	single
Legal	Getting assistance with legal issues	3	single
Basic Needs	Paying for basics like food, housing, medical care and heating; paying for food; getting or maintaining housing	3	composite
Work/Education	Finding or keeping work; getting additional education or job training	3	composite
I/L/D	Feeling socially isolated; feeling lonely; managing experiences of discrimination	3	composite
Mental Health	Q45: overall mental or emotional health rating	5	single
Well-being	Q46–48: percentage of time in past 4 weeks with good physical health, good emotional health, and ability to perform usual social activities	4	composite

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